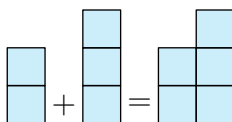


0.1

Addition with amounts

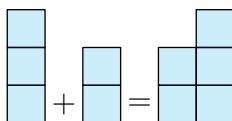
When we have an amount and wish to add more, we use the symbol $+$. If we have 2 and want to add 3, we write

$$2 + 3 = 5$$



The order in which we add have no impact on the results; starting off with 2 and adding 3 is the same as starting off with 3 and adding 2:

$$3 + 2 = 5$$



A calculation involving addition includes two or more *terms* and one *sum*. In the calculation

$$2 + 3 = 5$$

both 2 and 3 are terms while 5 is the sum.

Common ways of saying $2 + 3$ include

- "2 plus 3"
- "2 added to 3"
- "2 and 3 added"

0.1 Addition is commutative

The order of the terms has no impact on the sum.

$$2 + 5 = 7 = 5 + 2$$

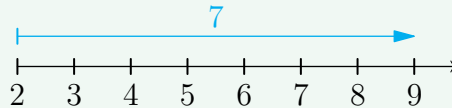
$$6 + 3 = 9 = 3 + 6$$

Addition on the number line: moving to the right

On a number line, addition with positive numbers involves moving *to the right*:

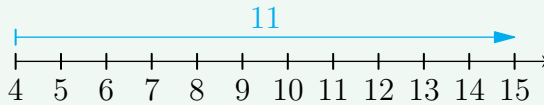
1

$$2 + 7 = 9$$



2

$$4 + 11 = 15$$



Interpretation of =

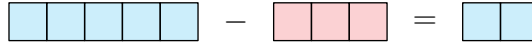
+ brings the possibility of expressing numbers in different ways, for example is $5 = 2 + 3$ and $5 = 1 + 4$. In this context, **=** means "has the same value as". This is also the case regarding subtraction, multiplication and division which we'll look at in the next three sections.

0.2

Subtraction with amounts

When removing a part of an amount, we use the symbol $-$:

$$5 - 3 = 2$$



A calculation involving subtraction includes one or more *terms* and one *difference*. In the calculation

$$5 - 3 = 2$$

both 5 and 3 are terms while 2 is the difference.

Common ways of saying $5 - 3$ include

- "5 minus 3"
- "3 subtracted from 5"

A new interpretation of 0

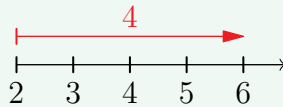
As mentioned earlier in this book, 0 can be interpreted as "nothing". However, subtraction brings the possibility of expressing 0 by other numbers, for example $7 - 7 = 0$ and $19 - 19 = 0$. In many practical situations, 0 indicates some form of equilibrium, like two equal but opposite forces.

Subtraction on the number line: Moving to the left

In Addition, we have seen that $+$ (with positive numbers) involves moving *to the right* on the number line. With $-$ it's the opposite, we move *to the left*¹:

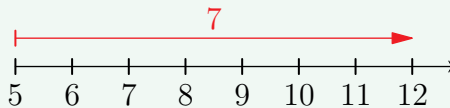
1

$$6 - 4 = 2$$



2

$$12 - 7 = 5$$



Notice

At first it may seem a bit odd moving in the opposite direction of the way in which the arrows point, as in *Example 1* and *2*. However, in Negtal this will turn out to be useful.

¹In figures with number lines, the red colored arrows indicates that you shall start at the arrowhead and move to the other end.

0.3

Multiplication by integers: initial definition

When adding equal numbers, we can use the multiplication symbol $\cdot$ to write our calculations more compact:

$$\begin{aligned}4 + 4 + 4 &= 4 \cdot 3 \\8 + 8 &= 8 \cdot 2 \\1 + 1 + 1 + 1 + 1 &= 1 \cdot 5\end{aligned}$$

A calculation involving multiplication includes several *factors* and one *product*. In the calculation

$$4 \cdot 3 = 12$$

both 4 and 3 are factors, while 12 is the product.

Common ways of saying $4 \cdot 3$ include

- "4 times 3"
- "4 multiplied by 3"
- "4 and 3 multiplied together"

A lot of texts use $\times$ instead of $\cdot$. In computer programming, $*$ is the most common symbol for multiplication.

Multiplication involving amounts

Let us illustrate $2 \cdot 3$:

$$2 \cdot 3 = \begin{array}{|c|} \hline \square \\ \hline \square \\ \hline \end{array} + \begin{array}{|c|} \hline \square \\ \hline \square \\ \hline \end{array} + \begin{array}{|c|} \hline \square \\ \hline \square \\ \hline \end{array} = \begin{array}{|c|c|c|} \hline \square & \square & \square \\ \hline \square & \square & \square \\ \hline \end{array}$$

Now notice the product of $3 \cdot 2$:

$$3 \cdot 2 = \begin{array}{|c|} \hline \square \\ \hline \square \\ \hline \square \\ \hline \end{array} + \begin{array}{|c|} \hline \square \\ \hline \square \\ \hline \square \\ \hline \end{array} = \begin{array}{|c|c|} \hline \square & \square \\ \hline \square & \square \\ \hline \square & \square \\ \hline \end{array}$$

0.2

The order of the factors has no impact on the product.

$$3 \cdot 4 = 12 = 4 \cdot 3$$

$$6 \cdot 7 = 42 = 7 \cdot 6$$

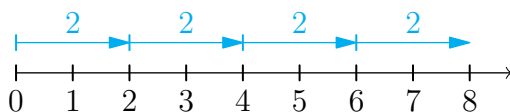
$$8 \cdot 9 = 72 = 9 \cdot 8$$

Multiplication on the number line

We can also use the number line to calculate multiplications. In the case of $2 \cdot 4$ we can think like this:

" $2 \cdot 4$ means moving 2 places to the right, 4 times."

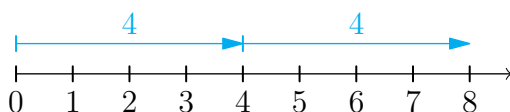
$$2 \cdot 4 = 8$$



We can also use the number line to prove to ourselves that multiplication is commutative:

" $4 \cdot 2$ means moving 4 places to the right, 2 times."

$$4 \cdot 2 = 8$$



Final definition of multiplication by positive integers

It may be the most intuitive to interpret " 2 times 3 " as " 3 , 2 times". Then it follows:

$$2 \text{ times } 3 = 3 + 3$$

In this section we introduced $2 \cdot 3$, that is " 2 times 3 ", as $2 + 2 + 2$. With this interpretation, $3 + 3$ corresponds to $3 \cdot 2$, but the fact that multiplication is a commutative operation (gangkom) ensures that the one interpretation does not exclude the other; $2 \cdot 3 = 2 + 2 + 2$ and $2 \cdot 3 = 3 + 3$ are two expressions of same value.

0.3 Multiplication as repeated addition

Multiplication involving a positive integer can be expressed as repeated addition.

1

$$4 + 4 + 4 = 4 \cdot 3 = 3 + 3 + 3 + 3$$

$$8 + 8 = 8 \cdot 2 = 2 + 2 + 2 + 2 + 2 + 2 + 2$$

$$1 + 1 + 1 + 1 + 1 = 1 \cdot 5 = 5$$

Notice

The fact that multiplication with positive integers can be expressed as repeated addition does not exclude other expressions. There's nothing wrong with writing $2 \cdot 3 = 1 + 5$.

0.4

$:$ is the symbol for division. Division has three different interpretations:

0.4 The three interpretations of division

- **Distribution of amounts**

$12 : 3 =$ "The number in each group when evenly distributing 12 into 3 groups"

- **Number of equal terms**

$12 : 3 =$ "The number of 3's added to make 12"

- **The inverse operation of multiplication**

$12 : 3 =$ "The number which yields 12 when multiplied by 3"

A calculation involving division includes a *dividend*, a *divisor* and a *quotient*. In the calculation

$$12 : 3 = 4$$

12 is the dividend, 3 is the divisor and 4 is the quotient.

Common ways of saying $12 : 3$ include

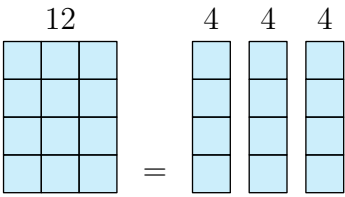
- "12 divided by 3"
- "12 to 3"

In a lot of contexts, $/$ is used instead of $:$, especially in computer programming.

Sometimes $12 : 3$ is called "the *ratio* of 12 to 3".

Distribution of amounts

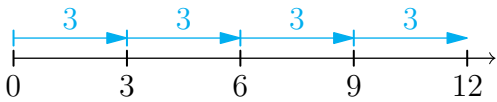
The calculation $12 : 3$ tells that we shall distribute 12 into 3 equal groups:



We observe that each group contains 4 boxes, which means that

$$12 : 3 = 4$$

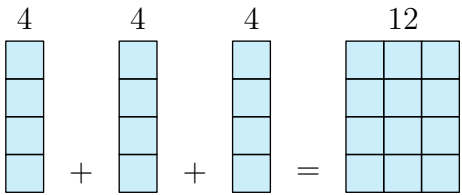
Number of equal terms



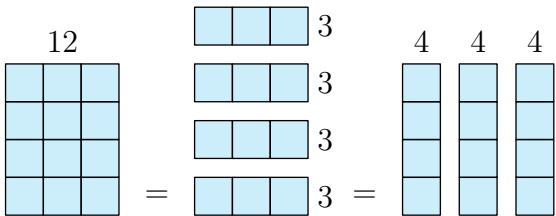
12 equals the sum of 4 instances of 3, that is $12 : 3 = 4$.

The inverse operation of multiplication

We have just seen that if we divide 12 into 3 equal groups, we get 4 in each group. Hence $12 : 3 = 4$. The sum of these groups makes 12:



However, this is the same as multiplying 4 by 3, in other words: If we know that $4 \cdot 3 = 12$, we also know that $12 : 3 = 4$. As well we know that $12 : 4 = 3$.



1

Since $6 \cdot 3 = 18$,

$$18 : 6 = 3$$

$$18 : 3 = 6$$

2

Since $5 \cdot 7 = 35$,

$$35 : 5 = 7$$

$$35 : 7 = 5$$